

of samples $x(1), x(2), \dots, x(n)$ of a random variable X will converge to $E[X]$ as n becomes large. For a discrete uniform $(0, 10)$ random variable X , use MATLAB to examine this convergence.

- (a) For 100 sample values of X , plot the sequence $m_1, m_2, \dots, m_{100}$. Repeat this experiment five times, plotting all five m_n curves on common axes.
- (b) Repeat part (a) for 1000 sample values of X .

Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

3.2.1 ● The random variable N has PMF

$$P_N(n) = \begin{cases} c(1/2)^n & n = 0, 1, 2, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is the value of the constant c ?
- (b) What is $P[N \leq 1]$?

3.2.2 ● The random variable V has PMF

$$P_V(v) = \begin{cases} cv^2 & v = 1, 2, 3, 4, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the value of the constant c .
- (b) Find $P[V \in \{u^2 | u = 1, 2, 3, \dots\}]$.
- (c) Find the probability that V is even.
- (d) Find $P[V > 2]$.

3.2.3 ● The random variable X has PMF

$$P_X(x) = \begin{cases} c/x & x = 2, 4, 8, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is the value of the constant c ?
- (b) What is $P[X = 4]$?
- (c) What is $P[X < 4]$?
- (d) What is $P[3 \leq X \leq 9]$?

3.2.4 ● In each at-bat in a baseball game, mighty Casey swings at every pitch. The

result is either a home run (with probability $q = 0.05$) or a strike. Of course, three strikes and Casey is out.

- (a) What is the probability $P[H]$ that Casey hits a home run?
- (b) For one at-bat, what is the PMF of N , the number of times Casey swings his bat?

3.2.5 ■ A tablet computer transmits a file over a wi-fi link to an access point. Depending on the size of the file, it is transmitted as N packets where N has PMF

$$P_N(n) = \begin{cases} c/n & n = 1, 2, 3, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the constant c .
- (b) What is the probability that N is odd?
- (c) Each packet is received correctly with probability p , and the file is received correctly if all N packets are received correctly. Find $P[C]$, the probability that the file is received correctly.

3.2.6 ■ In college basketball, when a player is fouled while not in the act of shooting and the opposing team is “in the penalty,” the player is awarded a “1 and 1.” In the 1 and 1, the player is awarded one free throw, and if that free throw goes in the player is awarded a second free throw. Find the PMF of Y , the number of points scored in

a 1 and 1 given that any free throw goes in with probability p , independent of any other free throw.

3.2.7 ■ You roll a 6-sided die repeatedly. Starting with roll $i = 1$, let R_i denote the result of roll i . If $R_i > i$, then you will roll again; otherwise you stop. Let N denote the number of rolls.

- (a) What is $P[N > 3]$?
- (b) Find the PMF of N .

3.2.8 ■ You are manager of a ticket agency that sells concert tickets. You assume that people will call three times in an attempt to buy tickets and then give up. You want to make sure that you are able to serve at least 95% of the people who want tickets. Let p be the probability that a caller gets through to your ticket agency. What is the minimum value of p necessary to meet your goal?

3.2.9 ■ In the ticket agency of Problem 3.2.8, each telephone ticket agent is available to receive a call with probability 0.2. If all agents are busy when someone calls, the caller hears a busy signal. What is the minimum number of agents that you have to hire to meet your goal of serving 95% of the customers who want tickets?

3.2.10 ■ Suppose when a baseball player gets a hit, a single is twice as likely as a double, which is twice as likely as a triple, which is twice as likely as a home run. Also, the player's batting average, i.e., the probability the player gets a hit, is 0.300. Let B denote the number of bases touched safely during an at-bat. For example, $B = 0$ when the player makes an out, $B = 1$ on a single, and so on. What is the PMF of B ?

3.2.11 ♦ When someone presses SEND on a cellular phone, the phone attempts to set up a call by transmitting a SETUP message to a nearby base station. The phone waits for a response, and if none arrives within 0.5 seconds it tries again. If it doesn't get a response after $n = 6$ tries, the phone stops transmitting messages and generates a busy signal.

- (a) Draw a tree diagram that describes the call setup procedure.
- (b) If all transmissions are independent and the probability is p that a SETUP message will get through, what is the PMF of K , the number of messages transmitted in a call attempt?
- (c) What is the probability that the phone will generate a busy signal?
- (d) As manager of a cellular phone system, you want the probability of a busy signal to be less than 0.02. If $p = 0.9$, what is the minimum value of n necessary to achieve your goal?

3.3.1 ● In a package of M&Ms, Y , the number of yellow M&Ms, is uniformly distributed between 5 and 15.

- (a) What is the PMF of Y ?
- (b) What is $P[Y < 10]$?
- (c) What is $P[Y > 12]$?
- (d) What is $P[8 \leq Y \leq 12]$?

3.3.2 ● In a bag of 25 M&Ms, each piece is equally likely to be red, green, orange, blue, or brown, independent of the color of any other piece. Find the PMF of R , the number of red pieces. What is the probability a bag has no red M&Ms?

3.3.3 ● When a conventional paging system transmits a message, the probability that the message will be received by the pager it is sent to is p . To be confident that a message is received at least once, a system transmits the message n times.

- (a) Assuming all transmissions are independent, what is the PMF of K , the number of times the pager receives the same message?
- (b) Assume $p = 0.8$. What is the minimum value of n that produces a probability of 0.95 of receiving the message at least once?

3.3.4 ● You roll a pair of fair dice until you roll "doubles" (i.e., both dice are the same). What is the expected number, $E[N]$, of rolls?

3.3.5 ● When you go fishing, you attach m hooks to your line. Each time you cast your line, each hook will be swallowed by a fish with probability h , independent of whether any other hook is swallowed. What is the PMF of K , the number of fish that are hooked on a single cast of the line?

3.3.6 ● Any time a child throws a Frisbee, the child's dog catches the Frisbee with probability p , independent of whether the Frisbee is caught on any previous throw. When the dog catches the Frisbee, it runs away with the Frisbee, never to be seen again. The child continues to throw the Frisbee until the dog catches it. Let X denote the number of times the Frisbee is thrown.

- What is the PMF $P_X(x)$?
- If $p = 0.2$, what is the probability that the child will throw the Frisbee more than four times?

3.3.7 ● When a two-way paging system transmits a message, the probability that the message will be received by the pager it is sent to is p . When the pager receives the message, it transmits an acknowledgment signal (ACK) to the paging system. If the paging system does not receive the ACK, it sends the message again.

- What is the PMF of N , the number of times the system sends the same message?
- The paging company wants to limit the number of times it has to send the same message. It has a goal of $P[N \leq 3] \geq 0.95$. What is the minimum value of p necessary to achieve the goal?

3.3.8 ● The number of bytes B in an HTML file is the geometric $(2.5 \cdot 10^{-5})$ random variable. What is the probability $P[B > 500,000]$ that a file has over 500,000 bytes?

3.3.9 ●

- Starting on day 1, you buy one lottery ticket each day. Each ticket is a winner with probability 0.1. Find the PMF of

K , the number of tickets you buy up to and including your fifth winning ticket.

- L is the number of flips of a fair coin up to and including the 33rd occurrence of tails. What is the PMF of L ?
- Starting on day 1, you buy one lottery ticket each day. Each ticket is a winner with probability 0.01. Let M equal the number of tickets you buy up to and including your first winning ticket. What is the PMF of M ?

3.3.10 ● The number of buses that arrive at a bus stop in T minutes is a Poisson random variable B with expected value $T/5$.

- What is the PMF of B , the number of buses that arrive in T minutes?
- What is the probability that in a two-minute interval, three buses will arrive?
- What is the probability of no buses arriving in a 10-minute interval?
- How much time should you allow so that with probability 0.99 at least one bus arrives?

3.3.11 ● In a wireless automatic meter-reading system, a base station sends out a wake-up signal to nearby electric meters. On hearing the wake-up signal, a meter transmits a message indicating the electric usage. Each message is repeated eight times.

- If a single transmission of a message is successful with probability p , what is the PMF of N , the number of successful message transmissions?
- I is an indicator random variable such that $I = 1$ if at least one message is transmitted successfully; otherwise $I = 0$. Find the PMF of I .

3.3.12 ● A Zipf $(n, \alpha = 1)$ random variable X has PMF

$$P_X(x) = \begin{cases} c(n)/x & x = 1, 2, \dots, n, \\ 0 & \text{otherwise.} \end{cases}$$

The constant $c(n)$ is set so that $\sum_{x=1}^n P_X(x) = 1$. Calculate $c(n)$ for $n = 1, 2, \dots, 6$.

3.3.13 ■ In a bag of 64 “holiday season” M&Ms, each M&M is equally likely to be red or green, independent of any other M&M in the bag.

- If you randomly grab four M&Ms, what is the probability $P[E]$ that you grab an equal number of red and green M&Ms?
- What is the PMF of G , the number of green M&Ms in the bag?
- You begin eating randomly chosen M&Ms one by one. Let R equal the number of red M&Ms you eat before you eat your first green M&M. What is the PMF of R ?

3.3.14 ■ A radio station gives a pair of concert tickets to the sixth caller who knows the birthday of the performer. For each person who calls, the probability is 0.75 of knowing the performer’s birthday. All calls are independent.

- What is the PMF of L , the number of calls necessary to find the winner?
- What is the probability of finding the winner on the tenth call?
- What is the probability that the station will need nine or more calls to find a winner?

3.3.15 ■ In a packet voice communications system, a source transmits packets containing digitized speech to a receiver. Because transmission errors occasionally occur, an acknowledgment (ACK) or a negative acknowledgment (NAK) is transmitted back to the source to indicate the status of each received packet. When the transmitter gets a NAK, the packet is retransmitted. Voice packets are delay sensitive, and a packet can be transmitted a maximum of d times. If a packet transmission is an independent Bernoulli trial with success probability p , what is the PMF of T , the number of times a packet is transmitted?

3.3.16 ♦ At Newark airport, your jet joins a line as the tenth jet waiting for takeoff. At Newark, takeoffs and landings are synchronized to the minute. In each one-minute

interval, an arriving jet lands with probability $p = 2/3$, independent of an arriving jet in any other minute. Such an arriving jet blocks any waiting jet from taking off in that one-minute interval. However, if there is no arrival, then the waiting jet at the head of the line takes off. Each take-off requires exactly one minute.

- Let L_1 denote the number of jets that land before the jet at the front of the line takes off. Find the PMF $P_{L_1}(l)$.
- Let W denote the number of minutes you wait until your jet takes off. Find $P[W = 10]$. (Note that if no jets land for ten minutes, then one waiting jet will take off each minute and $W = 10$.)
- What is the PMF of W ?

3.3.17 ♦ Suppose each day (starting on day 1) you buy one lottery ticket with probability $1/2$; otherwise, you buy no tickets. A ticket is a winner with probability p independent of the outcome of all other tickets. Let N_i be the event that on day i you do *not* buy a ticket. Let W_i be the event that on day i , you buy a winning ticket. Let L_i be the event that on day i you buy a losing ticket.

- What are $P[W_{33}]$, $P[L_{87}]$, and $P[N_{99}]$?
- Let K be the number of the day on which you buy your first lottery ticket. Find the PMF $P_K(k)$.
- Find the PMF of R , the number of losing lottery tickets you have purchased in m days.
- Let D be the number of the day on which you buy your j th losing ticket. What is $P_D(d)$? Hint: If you buy your j th losing ticket on day d , how many losers did you have after $d - 1$ days?

3.3.18 ♦ The Sixers and the Celtics play a best out of five playoff series. The series ends as soon as one of the teams has won three games. Assume that either team is equally likely to win any game independently of any other game played. Find

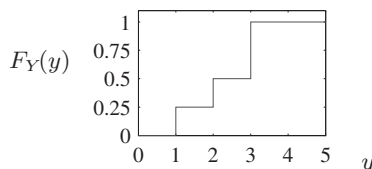
- The PMF $P_N(n)$ for the total number N of games played in the series;

- (b) The PMF $P_W(w)$ for the number W of Celtics wins in the series;
- (c) The PMF $P_L(l)$ for the number L of Celtics losses in the series.

3.3.19 ♦ For a binomial random variable K representing the number of successes in n trials, $\sum_{k=0}^n P_K(k) = 1$. Use this fact to prove the binomial theorem for any $a > 0$ and $b > 0$. That is, show that

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}.$$

3.4.1 ● Discrete random variable Y has the CDF $F_Y(y)$ as shown:



Use the CDF to find the following probabilities:

- (a) $P[Y < 1]$ and $P[Y \leq 1]$
- (b) $P[Y > 2]$ and $P[Y \geq 2]$
- (c) $P[Y = 3]$ and $P[Y > 3]$
- (d) $P_Y(y)$

3.4.2 ● The random variable X has CDF

$$F_X(x) = \begin{cases} 0 & x < -1, \\ 0.2 & -1 \leq x < 0, \\ 0.7 & 0 \leq x < 1, \\ 1 & x \geq 1. \end{cases}$$

- (a) Draw a graph of the CDF.
- (b) Write $P_X(x)$, the PMF of X . Be sure to write the value of $P_X(x)$ for all x from $-\infty$ to ∞ .

3.4.3 ● The random variable X has CDF

$$F_X(x) = \begin{cases} 0 & x < -3, \\ 0.4 & -3 \leq x < 5, \\ 0.8 & 5 \leq x < 7, \\ 1 & x \geq 7. \end{cases}$$

- (a) Draw a graph of the CDF.
- (b) Write $P_X(x)$, the PMF of X .

3.4.4 ● Following Example 3.22, show that a geometric (p) random variable K has CDF

$$F_K(k) = \begin{cases} 0 & k < 1, \\ 1 - (1 - p)^{\lfloor k \rfloor} & k \geq 1. \end{cases}$$

3.4.5 ■ At the One Top Pizza Shop, a pizza sold has mushrooms with probability $p = 2/3$. On a day in which 100 pizzas are sold, let N equal the number of pizzas sold before the first pizza with mushrooms is sold. What is the PMF of N ? What is the CDF of N ?

3.4.6 ■ In Problem 3.2.10, find and sketch the CDF of B , the number of bases touched safely during an at-bat.

3.4.7 ■ In Problem 3.2.6, find and sketch the CDF of Y , the number of points scored in a 1 and 1 for $p = 1/4$, $p = 1/2$, and $p = 3/4$.

3.4.8 ■ In Problem 3.2.11, find and sketch the CDF of N , the number of attempts made by the cellular phone for $p = 1/2$.

3.5.1 ● Let X have the uniform PMF

$$P_X(x) = \begin{cases} 0.01 & x = 1, 2, \dots, 100, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find a mode x_{mod} of X . If the mode is not unique, find the set X_{mod} of all modes of X .
- (b) Find a median x_{med} of X . If the median is not unique, find the set X_{med} of all numbers x that are medians of X .

3.5.2 ● It costs 20 cents to receive a photo and 30 cents to send a photo from a cell-phone. C is the cost of one photo (either sent or received). The probability of receiving a photo is 0.6. The probability sending a photo is 0.4.

- (a) Find $P_C(c)$, the PMF of C .

(b) What is $E[C]$, the expected value of C ?

3.5.3 ●

- (a) The number of trains J that arrive at the station in time t minutes is a Poisson random variable with $E[J] = t$. Find t such that $P[J > 0] = 0.9$.
- (b) The number of buses K that arrive at the station in one hour is a Poisson random variable with $E[K] = 10$. Find $P[K = 10]$.
- (c) In a 1 ms interval, the number of hits L on a Web server is a Poisson random variable with expected value $E[L] = 2$ hits. What is $P[L \leq 1]$?

3.5.4 ● You simultaneously flip a pair of fair coins. Your friend gives you one dollar if both coins come up heads. You repeat this ten times and your friend gives you X dollars. Find $E[X]$, the expected number of dollars you receive. What is the probability that you do “worse than average”?

3.5.5 ● A packet received by your smartphone is error-free with probability 0.95, independent of any other packet.

- (a) Out of 10 packets received, let X equal the number of packets received with errors. What is the PMF of X ?
- (b) In one hour, your smartphone receives 12,000 packets. Let X equal the number of packets received with errors. What is $E[X]$?

3.5.6 ● Find the expected value of the random variable Y in Problem 3.4.1.

3.5.7 ● Find the expected value of the random variable X in Problem 3.4.2.

3.5.8 ● Find the expected value of the random variable X in Problem 3.4.3.

3.5.9 ● Use Definition 3.13 to calculate the expected value of a binomial $(4, 1/2)$ random variable X .

3.5.10 ● X is the discrete uniform $(1, 5)$ random variable.

- (a) What is $P[X = E[X]]$?

(b) What is $P[X > E[X]]$?

3.5.11 ● K is the geometric $(1/11)$ random variable.

- (a) What is $P[K = E[K]]$?
- (b) What is $P[K > E[K]]$?
- (c) What is $P[K < E[K]]$?

3.5.12 ● At a casino, people line up to pay \$20 each to be a contestant in the following game: The contestant flips a fair coin repeatedly. If she flips heads 20 times in a row, she walks away with $R = 20$ million dollars; otherwise she walks away with $R = 0$ dollars.

- (a) Find the PMF of R , the reward earned by the contestant.
- (b) The casino counts “losing contestants” who fail to win the 20 million dollar prize. Let L equal the number of losing contestants before the first winning contestant. What is the PMF of L ?
- (c) Why does the casino offer this game?

3.5.13 ● Give examples of practical applications of probability theory that can be modeled by the following PMFs. In each case, state an experiment, the sample space, the range of the random variable, the PMF of the random variable, and the expected value:

- (a) Bernoulli
- (b) Binomial
- (c) Pascal
- (d) Poisson

Make up your own examples. (Don’t copy examples from the text.)

3.5.14 ■ Find $P[K < E[K]]$ when

- (a) K is geometric $(1/3)$.
- (b) K is binomial $(6, 1/2)$.
- (c) K is Poisson (3) .
- (d) K is discrete uniform $(0, 6)$.

3.5.15 ■ Suppose you go to a casino with exactly \$63. At this casino, the only game is roulette and the only bets allowed are red and green. The payoff for a winning bet

is the amount of the bet. In addition, the wheel is fair so that $P[\text{red}] = P[\text{green}] = 1/2$. You have the following strategy: First, you bet \$1. If you win the bet, you quit and leave the casino with \$64. If you lose, you then bet \$2. If you win, you quit and go home. If you lose, you bet \$4. In fact, whenever you lose, you double your bet until either you win a bet or you lose all of your money. However, as soon as you win a bet, you quit and go home. Let Y equal the amount of money that you take home. Find $P_Y(y)$ and $E[Y]$. Would you like to play this game every day?

3.5.16 ♦ In a TV game show, there are three identical-looking suitcases. The first suitcase has 3 dollars, the second has 30 dollars and the third has 300 dollars. You start the game by randomly choosing a suitcase. *Between the two unchosen suitcases*, the game show host opens the suitcase with more money. The host then asks you if you want to keep your suitcase or switch to the other remaining suitcase. After you make your decision, you open your suitcase and keep the D dollars inside. Should you switch suitcases? To answer this question, solve the following subproblems and use the following notation:

- C_i is the event that you first choose the suitcase with i dollars.
- O_i denotes the event that the host opens a suitcase with i dollars.

In addition, you may wish to go back and review the Monty Hall problem in Example 2.4.

- (a) Suppose you never switch; you always stick with your original choice. Use a tree diagram to find the PMF $P_D(d)$ and expected value $E[D]$.
- (b) Suppose you always switch. Use a tree diagram to find the PMF $P_D(d)$ and expected value $E[D]$.
- (c) Perhaps your rule for switching should depend on how many dollars are in the suitcase that the host opens? What is the optimal strategy to maximize

$E[D]$? Hint: Consider making a random decision; if the host opens a suitcase with i dollars, let α_i denote the probability that you switch.

3.5.17 ♦ You are a contestant on a TV game show; there are four identical-looking suitcases containing \$100, \$200, \$400, and \$800. You start the game by randomly choosing a suitcase. *Among the three unchosen suitcases*, the game show host opens the suitcase that holds the median amount of money. (For example, if the unopened suitcases contain \$100, \$400 and \$800, the host opens the \$400 suitcase.) The host then asks you if you want to keep your suitcase or switch one of the other remaining suitcases. For your analysis, use the following notation for events:

- C_i is the event that you choose a suitcase with i dollars.
- O_i denotes the event that the host opens a suitcase with i dollars.
- R is the reward in dollars that you keep.

- (a) You refuse the host's offer and open the suitcase you first chose. Find the PMF of R and the expected value $E[R]$.
- (b) You always switch and randomly choose one of the two remaining suitcases with equal probability. You receive the R dollars in this chosen suitcase. Sketch a tree diagram for this experiment, and find the PMF and expected value of R .
- (c) Can you do better than either always switching or always staying with your original choice? Explain.

3.5.18 ♦ You are a contestant on a TV game show; there are four identical-looking suitcases containing \$200, \$400, \$800, and \$1600. You start the game by randomly choosing a suitcase. *Among the three unchosen suitcases*, the game show host opens the suitcase that holds the least money. The host then asks you if you want to keep

your suitcase or switch one of the other remaining suitcases. For the following analysis, use the following notation for events:

- C_i is the event that you choose a suitcase with i dollars.
 - O_i denotes the event that the host opens a suitcase with i dollars.
 - R is the reward in dollars that you keep.
- (a) You refuse the host's offer and open the suitcase you first chose. Find the PMF of R and the expected value $E[R]$.
- (b) You switch and randomly choose one of the two remaining suitcases. You receive the R dollars in this chosen suitcase. Sketch a tree diagram for this experiment, and find the PMF and expected value of R .

3.5.19 ♦ Let binomial random variable X_n denote the number of successes in n Bernoulli trials with success probability p . Prove that $E[X_n] = np$. Hint: Use the fact that $\sum_{x=0}^{n-1} P_{X_{n-1}}(x) = 1$.

3.5.20 ♦ Prove that if X is a nonnegative integer-valued random variable, then

$$E[X] = \sum_{k=0}^{\infty} P[X > k].$$

3.5.21 ♦ At the gym, a weightlifter can bench press a maximum of 100 kg. For a mass of m kg, ($1 \leq m \leq 100$), the maximum number of repetitions she can complete is R , a geometric random variable with expected value $100/m$.

- (a) In terms of the mass m , what is the PMF of R ?
- (b) When she performs one repetition, she lifts the m kg mass a height $h = 5/9.8$ meters and thus does work $w = mgh = 5m$ Joules. For R repetitions, she does $W = 5mR$ Joules of work. What is the expected work $E[W]$ that she will complete?
- (c) A friend offers to pay her 1000 dollars if she can perform 1000 Joules of

weightlifting work. What mass m in the range $1 \leq m \leq 100$ should she use to maximize her probability of winning the money? For the best choice of m , what is the probability that she wins the money?

3.5.22 ♦ At the gym, a weightlifter can bench press a maximum of 200 kg. For a mass of m kg, ($1 \leq m \leq 200$), the maximum number of repetitions she can complete is R , a geometric random variable with expected value $200/m$.

- (a) In terms of the mass m , what is the PMF of R ?
- (b) When she performs one repetition, she lifts the m kg mass a height $h = 4/9.8$ meters and thus does work $w = mgh = 4m$ Joules. For R repetitions, she does $W = 4mR$ Joules of work. What is the expected work $E[W]$ that she will complete?
- (c) A friend offers to pay her 1000 dollars if she can perform 1000 Joules of weightlifting work. What mass m in the range $1 \leq m \leq 200$ should she use to maximize her probability of winning the money?

3.6.1 ● Given the random variable Y in Problem 3.4.1, let $U = g(Y) = Y^2$.

- (a) Find $P_U(u)$.
- (b) Find $F_U(u)$.
- (c) Find $E[U]$.

3.6.2 ● Given the random variable X in Problem 3.4.2, let $V = g(X) = |X|$.

- (a) Find $P_V(v)$.
- (b) Find $F_V(v)$.
- (c) Find $E[V]$.

3.6.3 ● Given the random variable X in Problem 3.4.3, let $W = g(X) = -X$.

- (a) Find $P_W(w)$.
- (b) Find $F_W(w)$.
- (c) Find $E[W]$.

3.6.4 ● At a discount brokerage, a stock purchase or sale worth less than \$10,000 incurs a brokerage fee of 1% of the value of the transaction. A transaction worth more than \$10,000 incurs a fee of \$100 plus 0.5% of the amount exceeding \$10,000. Note that for a fraction of a cent, the brokerage always charges the customer a full penny. You wish to buy 100 shares of a stock whose price D in dollars has PMF

$$P_D(d) = \begin{cases} 1/3 & d = 99.75, 100, 100.25, \\ 0 & \text{otherwise.} \end{cases}$$

What is the PMF of C , the cost of buying the stock (including the brokerage fee)?

3.6.5 ■ A source transmits data packets to a receiver over a radio link. The receiver uses error detection to identify packets that have been corrupted by radio noise. When a packet is received error free, the receiver sends an acknowledgment (ACK) back to the source. When the receiver gets a packet with errors, a negative acknowledgment (NAK) message is sent back to the source. Each time the source receives a NAK, the packet is retransmitted. We assume that each packet transmission is independently corrupted by errors with probability q .

- Find the PMF of X , the number of times that a packet is transmitted by the source.
- Suppose each packet takes 1 millisecond to transmit and that the source waits an additional millisecond to receive the acknowledgment message (ACK or NAK) before retransmitting. Let T equal the time required until the packet is successfully received. What is the relationship between T and X ? What is the PMF of T ?

3.6.6 ■ Suppose that a cellular phone costs \$20 per month with 30 minutes of use included and that each additional minute of use costs \$0.50. If the number of minutes you use in a month is a geometric random variable M with expected value of

$E[M] = 1/p = 30$ minutes, what is the PMF of C , the cost of the phone for one month?

3.6.7 ■ A professor tries to count the number of students attending lecture. For each student in the audience, the professor either counts the student properly (with probability p) or overlooks (and does not count) the student with probability $1 - p$. The exact number of attending students is 70.

- The number of students counted by the professor is a random variable N . What is the PMF of N ?
- Let $U = 70 - N$ denote the number of uncounted students. What is the PMF of U ?
- What is the probability that the undercount U is 2 or more?
- For what value of p does $E[U] = 2$?

3.6.8 ■ A forgetful professor tries to count the M&Ms in a package; however, the professor often loses his place and double counts an M&M. For each M&M in the package, the professor counts the M&M and then, with probability p counts the M&M again. The exact number of M&Ms in the pack is 20.

- Find the PMF of R , the number of double-counted M&Ms.
- Find the PMF of N , the number of M&Ms counted by the professor.

3.7.1 ● Starting on day $n = 1$, you buy one lottery ticket each day. Each ticket costs 1 dollar and is independently a winner that can be cashed for 5 dollars with probability 0.1; otherwise the ticket is worthless. Let X_n equal your net profit after n days. What is $E[X_n]$?

3.7.2 ● For random variable T in Quiz 3.6, first find the expected value $E[T]$ using Theorem 3.10. Next, find $E[T]$ using Definition 3.13.

3.7.3 ● In a certain lottery game, the chance of getting a winning ticket is exactly one in a thousand. Suppose a person buys one ticket each day (except on the leap year day February 29) over a period of fifty years.

What is the expected number $E[T]$ of winning tickets in fifty years? If each winning ticket is worth \$1000, what is the expected amount $E[R]$ collected on these winning tickets? Lastly, if each ticket costs \$2, what is your expected net profit $E[Q]$?

3.7.4 ● Suppose an NBA basketball player shooting an uncontested 2-point shot will make the basket with probability 0.6. However, if you foul the shooter, the shot will be missed, but two free throws will be awarded. Each free throw is an independent Bernoulli trial with success probability p . Based on the expected number of points the shooter will score, for what values of p may it be desirable to foul the shooter?

3.7.5 ● It can take up to four days after you call for service to get your computer repaired. The computer company charges for repairs according to how long you have to wait. The number of days D until the service technician arrives and the service charge C , in dollars, are described by

d	1	2	3	4
$P_D(d)$	0.2	0.4	0.3	0.1

and

$$C = \begin{cases} 90 & \text{for 1-day service,} \\ 70 & \text{for 2-day service,} \\ 40 & \text{for 3-day service,} \\ 40 & \text{for 4-day service.} \end{cases}$$

- What is the expected waiting time $\mu_D = E[D]$?
- What is the expected deviation $E[D - \mu_D]$?
- Express C as a function of D .
- What is the expected value $E[C]$?

3.7.6 ■ True or False: For any random variable X , $E[1/X] = 1/E[X]$.

3.7.7 ■ For the cellular phone in Problem 3.6.6, express the monthly cost C as a function of M , the number of minutes used. What is the expected monthly cost $E[C]$?

3.7.8 ■ A new cellular phone billing plan costs \$15 per month plus \$1 for each minute of use. If the number of minutes you use the phone in a month is a geometric random variable with expected value $1/p$, what is the expected monthly cost $E[C]$ of the phone? For what values of p is this billing plan preferable to the billing plan of Problem 3.6.6 and Problem 3.7.7?

3.7.9 ♦ A particular circuit works if all 10 of its component devices work. Each circuit is tested before leaving the factory. Each working circuit can be sold for k dollars, but each nonworking circuit is worthless and must be thrown away. Each circuit can be built with either ordinary devices or ultra-reliable devices. An ordinary device has a failure probability of $q = 0.1$ and costs \$1. An ultrareliable device has a failure probability of $q/2$ but costs \$3. Assuming device failures are independent, should you build your circuit with ordinary devices or ultra-reliable devices in order to maximize your expected profit $E[R]$? Keep in mind that your answer will depend on k .

3.7.10 ♦♦ In the New Jersey state lottery, each \$1 ticket has six randomly marked numbers out of $1, \dots, 46$. A ticket is a winner if the six marked numbers match six numbers drawn at random at the end of a week. For each ticket sold, 50 cents is added to the pot for the winners. If there are k winning tickets, the pot is divided equally among the k winners. Suppose you bought a winning ticket in a week in which $2n$ tickets are sold and the pot is n dollars.

- What is the probability q that a random ticket will be a winner?
- Find the PMF of K_n , the number of other (besides your own) winning tickets.
- What is the expected value of W_n , the prize for your winning ticket?

3.7.11 ♦♦ If there is no winner for the lottery described in Problem 3.7.10, then the pot is carried over to the next week. Suppose that in a given week, an r dollar pot is carried over from the previous week and

$2n$ tickets sold. Answer the following questions.

- What is the probability q that a random ticket will be a winner?
- If you own one of the $2n$ tickets sold, what is the expected value of V , the value (i.e., the amount you win) of that ticket? Is it ever possible that $E[V] > 1$?
- Suppose that in the instant before the ticket sales are stopped, you are given the opportunity to buy one of each possible ticket. For what values (if any) of n and r should you do it?

3.8.1 In an experiment to monitor two packets, the PMF of N , the number of video packets, is

n	0	1	2
$P_N(n)$	0.2	0.7	0.1

Find $E[N]$, $E[N^2]$, $\text{Var}[N]$, and σ_N .

3.8.2 Find the variance of the random variable Y in Problem 3.4.1.

3.8.3 Find the variance of the random variable X in Problem 3.4.2.

3.8.4 Find the variance of the random variable X in Problem 3.4.3.

3.8.5 Let X have the binomial PMF

$$P_X(x) = \binom{4}{x} (1/2)^4.$$

- Find the standard deviation of X .
- What is $P[\mu_X - \sigma_X \leq X \leq \mu_X + \sigma_X]$, the probability that X is within one standard deviation of the expected value?

3.8.6 X is the binomial $(5, 0.5)$ random variable.

- Find the standard deviation of X .
- Find $P[\mu_X - \sigma_X \leq X \leq \mu_X + \sigma_X]$, the probability that X is within one standard deviation of the expected value.

3.8.7 Show that the variance of $Y = aX + b$ is $\text{Var}[Y] = a^2 \text{Var}[X]$.

3.8.8 Given a random variable X with expected value μ_X and variance σ_X^2 , find the expected value and variance of

$$Y = \frac{X - \mu_X}{\sigma_X}.$$

3.8.9 In real-time packet data transmission, the time between successfully received packets is called the *interarrival time*, and randomness in packet interarrival times is called *jitter*. Jitter is undesirable. One measure of jitter is the standard deviation of the packet interarrival time. From Problem 3.6.5, calculate the jitter σ_T . How large must the successful transmission probability q be to ensure that the jitter is less than 2 milliseconds?

3.8.10 Random variable K has a Poisson (α) distribution. Derive the properties $E[K] = \text{Var}[K] = \alpha$. Hint: $E[K^2] = E[K(K-1)] + E[K]$.

3.8.11 For the delay D in Problem 3.7.5, what is the standard deviation σ_D of the waiting time?

3.9.1 Let X be the binomial $(100, 1/2)$ random variable. Let E_2 denote the event that X is a perfect square. Calculate $P[E_2]$.

3.9.2 Write a MATLAB function `x=shipweight8(m)` that produces m random sample values of the package weight X with PMF given in Example 3.27.

3.9.3 Use the `unique` function to write a MATLAB script `shipcostpmf.m` that outputs the pair of vectors `sy` and `py` representing the PMF $P_Y(y)$ of the shipping cost Y in Example 3.27.

3.9.4 For $m = 10$, $m = 100$, and $m = 1000$, use MATLAB to find the average cost of sending m packages using the model of Example 3.27. Your program input should have the number of trials m as the input. The output should be $\bar{Y} = \frac{1}{m} \sum_{i=1}^m Y_i$, where Y_i is the cost of the i th package. As m becomes large, what do you observe?

3.9.5 ■ The Zipf $(n, \alpha = 1)$ random variable X introduced in Problem 3.3.12 is often used to model the “popularity” of a collection of n objects. For example, a Web server can deliver one of n Web pages. The pages are numbered such that the page 1 is the most requested page, page 2 is the second most requested page, and so on. If page k is requested, then $X = k$.

To reduce external network traffic, an ISP gateway caches copies of the k most popular pages. Calculate, as a function of n for $1 \leq n \leq 1000$, how large k must be to ensure that the cache can deliver a page with probability 0.75.

3.9.6 ■ Generate n independent samples of the Poisson (5) random variable Y . For each $y \in S_Y$, let $n(y)$ denote the number of times that y was observed. Thus $\sum_{y \in S_Y} n(y) = n$ and the relative frequency of y is $R(y) = n(y)/n$. Compare the relative frequency of y against $P_Y(y)$ by plotting $R(y)$ and $P_Y(y)$ on the same graph as functions of y for $n = 100$, $n = 1000$ and

$n = 10,000$. How large should n be to have reasonable agreement?

3.9.7 ■ Test the convergence of Theorem 3.8. For $\alpha = 10$, plot the PMF of K_n for $(n, p) = (10, 1)$, $(n, p) = (100, 0.1)$, and $(n, p) = (1000, 0.01)$ and compare each result with the Poisson (α) PMF.

3.9.8 ■ Use the result of Problem 3.4.4 and the Random Sample Algorithm on Page 102 to write a MATLAB function `k=geometricrv(p,m)` that generates m samples of a geometric (p) random variable.

3.9.9 ♦ Find n^* , the smallest value of n for which the function `poissonpmf(n,n)` shown in Example 3.37 reports an error. What is the source of the error? Write a function `bigpoissonpmf(alpha,n)` that calculates `poissonpmf(n,n)` for values of n much larger than n^* . Hint: For a Poisson (α) random variable K ,

$$P_K(k) = \exp \left(-\alpha + k \ln(\alpha) - \sum_{j=1}^k \ln(j) \right).$$